Is It Spacetime?

Discrete universes measured by counting: a manifold detector, a first particle, and the memory a growing universe needs

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Not for circulation. Claims are tagged established, literature or ours. Table 2 lists what is ours after a literature crawl.

Abstract

We model spacetime as a discrete set of events with a causal order and nothing else (causal set theory), and ask two questions: which ways of producing such universes give something that looks like spacetime, and what a growing universe has to remember. Most of the machinery is known; we say where. *Part I.* (1) With Sorkin, we take the age of the universe as the infrared scale that tames the non-locality of Lorentz-invariant discreteness; it caps the rapidity of links at $\eta_{\max} \approx 142$, and a finite past leaves Sorkin's $1+1$ d'Alembertian exact up to e^{-V}. (2) A $1+1$ counting toolkit: an exact rectangle identity $d^2 = 2V_A V_B / V_C$ that extends Boguñá–Krioukov's overlap distance to any reference event, curvature from how that estimate drifts with the depth of the reference event $(1 - R\tau^2/16)$, and calibrated statistics for a Lorentzian-embedding manifold test in the spirit of Clough and Evans. (3) Applied to classical growth (fails), $e^{-\beta S}$ weights (the known continuum and layered phases; at $N = 400$ hot and cold chains never meet over $0.08 \le \beta \le 0.7$, consistent with the known first-order transition) and $e^{i\beta S}$ weights (few-layer orders are suppressed, but at reachable β only through the spread of the action). (4) Johnston's chain-sum particle: at fixed interval size n in a random 2D order its mean is exactly $\frac{1}{2} L_n(\mu/2)$, a known combinatorial identity not previously applied to the propagator; it separates the phases. *Part II.* (5) Quantum growth with one amplitude per birth needs no normalization; we propose that the past is stable only if the "future factors" of stems converge in ratio, and compute them exactly to $N = 11$: the smallest stem converges inside the known suppression window, larger stems do not yet. (6) Memory: each birth needs only a few past events, but spread over the whole age; per de Sitter patch the memory is a spacetime-volume law (Little's law), the counting behind Sorkin's $N \sim 10^{240}$ versus $\Lambda \sim N^{-1/2}$. Neither coarse cells nor endpoint phases reduce it; in small exact sums, phases make records rather than save memory. Gravity in $3+1$ and interactions are absent.

1 The question and the rules

In 1900 Planck assumed that energy comes in discrete quanta. We ask the analogous question for spacetime: can it be a discrete set of events that only *looks* continuous from far away, with general relativity (GR) as its large-scale description? The working constraints are:

C1 Discrete spacetime. Events and the causal order between them; the continuum is a large-scale limit.

C2 No background. No absolute space and no fixed aether; every statement is about relations between events.

C3 Non-locality regulated by the finite size of the universe (Sec. 2).

C4 Local Lorentz invariance for all boosts well below the cutoff.

C5 A cosmological clock computed from the universe's own state, e.g. the longest chain back to the beginning. It may be used as a bookkeeping clock for growth, but local laws must not depend on it, and predictions must not depend on which relational clock is used. Its frame should coincide with the matter (CMB) frame established; that is a test, not an input.

C6 Newton and Einstein from far away.

C7 Creation (and perhaps destruction) of events drives expansion, possibly through quantum creation operators.

C8 The links between events are quantum.

Missing so far is *matter*: nothing in the model yet plays the role of what gravitates.

A *fixed background* is structure the laws refer to but which nothing affects; C2 forbids it. A *dynamical preferred-frame field*, such as the Einstein-aether [1], breaks local Lorentz invariance. A *preferred frame of the state*, such as the CMB frame, comes from the contents of the universe while the laws stay invariant. C5 allows only the last kind.

The storyline. *Part I, "is it spacetime?"*: Section 2 explains why the finite size of the universe is our regulator; Sec. 3 builds tools that answer "is it spacetime?" from the order alone; Sec. 4 applies them to classical growth and to classical and quantum weights; Sec. 5 puts a first particle on the order. *Part II, "what must a growing universe remember?"*: Sec. 6 asks whether quantum growth has a stable past, and Sec. 7 how much of the past a growing universe must store, classically and quantum mechanically. The two parts share the growth step (11) and could become two notes (Sec. 8).

Part I: is it spacetime?

2 Neighbours and the cosmological cutoff

One neighbour per unit of rapidity. literature Following causal set theory [2, 3], y is a *neighbour* (a *link*) of x if y is in the causal future of x and no other event lies causally between them. In $1+1$ dimensions, with density ρ, write a future event as $u = t - x = \tau e^{-\eta}$, $v = t + x = \tau e^{\eta}$, where τ is the proper time from the origin and η the rapidity. The area element is $dt\,dx = \tau\,d\tau\,d\eta$, and the interval up to the event has area $\tau^2/2$, so it is empty with probability $e^{-\rho\tau^2/2}$. Therefore

$$\langle N_{\text{links}} \rangle = \int d\eta \int_0^\infty \rho\,\tau\,e^{-\rho\tau^2/2}\,d\tau = \int d\eta \cdot 1. \tag{1}$$

There is one expected neighbour per unit of rapidity, so an infinite universe gives every event infinitely many neighbours. This is the elementary core of the Bombelli–Henson–Sorkin theorem [4]: Lorentz-invariant discreteness is non-local [5].

A finite universe caps the rapidity. literature Sorkin already noted that recovering locality needs "an infrared scale R, which we may identify with the age of the cosmos", and mentioned a cosmic rest frame [5]; a finite rapidity cutoff singles out a frame, and the expected $1+1$ link count grows linearly with it [43]. ours We make the cap explicit. A neighbour at proper time τ and rapidity η lies at distance $r \approx (\tau/2)e^{\eta}$. In a universe of size L this cannot exceed L, so

$$\eta_{\max} \simeq \ln \frac{2L}{\ell_{\text{P}}} \approx 142, \tag{2}$$

with $L \approx 4.4 \times 10^{26}\,\mathrm{m}$ and $\ell_\mathrm{P} = 1.6 \times 10^{-35}\,\mathrm{m}$. Lorentz kinematics has been tested up to $\eta \approx 25$–27 (GZK protons and the highest-energy cosmic rays [6]), far below the cutoff. The only frame singled out is that of the universe as a whole, a frame of the state, as C5 requires. In $1+1$ dimensions an event has about $2\eta_\mathrm{max} \approx 280$ future neighbours. In $3+1$ dimensions the cutoff leaves $\sim (L/\ell_\mathrm{P})^2 \approx 10^{124}$ (Appendix A lists the numerology we avoid).

Does the cutoff show up in wave propagation? literature Sorkin's $1+1$ d'Alembertian [5] acts on a field by summing it over the past of x with layer weights. For density $\rho = \ell^{-2} = 1$ and averaged over sprinklings,

$$\bar{B}\phi(x) = 4\Big[-\tfrac{1}{2}\phi(x) + \int_{J^-(x)} d^2y\; e^{-V}\big(1 - 2V + \tfrac{1}{2}V^2\big)\phi(y)\Big], \tag{3}$$

where V is the volume of the interval between y and x. The three terms are links, next-to-links and next-to-next-to-links with weights $1, -2, 1$. On a plane wave $\phi = e^{-i(pu+qv)}$ it gives $4pq$, which is $\Box\phi/\phi$ in the convention used here (checked numerically).

literature / ours The kernel integrates to *zero* per unit rapidity, $\int_0^\infty \tau\, d\tau\, e^{-\tau^2/2}(1-\tau^2+\tau^4/8) = 0$, but there are infinitely many units of rapidity. The integral in Eq. (3) is therefore only conditionally convergent: when ϕ does not oscillate along one light-cone direction, the answer depends on how the far past is cut off. We compared two cut-offs in the cosmological frame.

1. *A rapidity wedge*, $|\eta| \le \eta_\mathrm{max}$, with τ integrated to infinity. Write the past event as $(u, v) = -(\tau e^{-\eta}, \tau e^{\eta})$. Then $s = pe^{-\eta} + qe^{\eta}$ is the phase per unit τ, and the bracket becomes $-\frac{1}{2} + \int d\eta\, h(s)$, with h the τ-integral of the kernel times $e^{is\tau}$.
2. *A finite past region*, the diamond $0 \le a \le U$, $0 \le b \le U$ of displacements along u and v ($U \simeq L$). This is what a universe of finite age provides. The b-integral is elementary (incomplete gamma functions), and a is integrated numerically.

For a constant field, the diamond gives the bracket in closed form: $-\frac{1}{2} + \frac{1}{2}\big[1-(1-A)e^{-A}\big]$ with $A = U^2/2$, i.e. $\langle B1(x)\rangle = -2(1-A)e^{-A}$, zero up to e^{-A}. literature This is the per-event form of a known result. The 2D action is $S = -\sum_x B1(x)$, and every event of an interval sees the sub-interval below it as its past, so $\langle S\rangle = 2\rho\int(1-A_x)e^{-A_x}\,d^2x = 2(1-e^{-A}) \to 2$: the volume of the interval's codimension-2 corner (its two side points) found by Buck, Dowker, Jubb and Surya [44]. We checked this by quadrature and on sprinklings (`sims/r7_corner_check.py`). Where in the region the compensating $\frac{1}{2}$ "comes from" is not meaningful, since the integral converges only conditionally. The results:

field	wedge	finite past	$\Box$
constant	-2	0	0
exact null wave ($p = 0$)	-1	0	0
massive, $pq = 10^{-9}$, a light-cone wavelength $> L$ ($L = e^{10}$)	-0.6 to -1.8	$4pq$	$4pq$

With an oscillating field the missing $\frac{1}{4}$ per null direction comes from the interior, at $u \sim 1/p$; with a non-oscillating one it comes from the edge of the past. The wedge has no edge, so it loses it. Boguñá and Krioukov found the same -2 for a constant field by integrating over rapidity first, obtained 0 with a spatial cut, and stressed that the limits do not commute [42]; for a constant field the finite past is the per-event version of [44]. What we add is only the reading: the finite age of the universe is the physical regulator, and with it the operator stays exact up to e^{-A} for every wavelength, including light-cone wavelengths longer than the universe, in agreement with the analytic continuation of [23]. Aslanbeigi, Saravani and Sorkin [23] define the plane-wave eigenvalue only where the integral converges absolutely (complex momenta with a timelike, future-pointing imaginary part) and reach real momenta as a limit. Their $1+1$

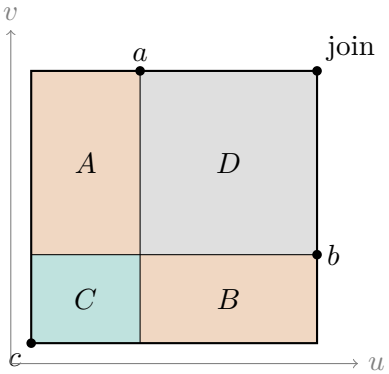


Figure 1: The rectangle identity in light-cone coordinates (rotate by 45° for the usual picture). The rectangle from c to the join of a and b splits into four tiles: the overlap C, the strips A and B seen by only one of a, b, and the diamond D with a and b at its corners. Opposite tiles have equal products, $C \cdot D = A \cdot B$, and D encodes the distance between a and b.

result, $g(Z) = -Ze^{Z/2}E_2(Z/2)$ with $Z = p \cdot p/\rho$, goes smoothly to zero at null momentum, in agreement with the finite past.

Consequence. In $1+1$ the finite size of the universe leaves Sorkin's $\Box$ exactly continuum-like, up to $e^{-(\text{past volume})}$, for every wavelength. Through this operator the cutoff produces no dispersion correction and no change in the speed of light. That is good news for C3 and C4, which do not conflict here, and bad news for predictions. The $3+1$ case is not done.

3 Geometry by counting

Everything in this section uses only the causal order. To test the tools we generate universes by Poisson sprinkling at density ρ into a known spacetime, discard the coordinates, and compare what the tools find with the truth. We write $\ell = \rho^{-1/2}$ for the discreteness length in $1+1$ dimensions: a patch of volume ℓ^2 holds one event on average.

3.1 Time: the longest chain

literature The proper time between two events is estimated by the longest chain between them [7]. In a sprinkled 1+1 diamond this is the longest increasing subsequence of a random permutation, with mean $\simeq 2\sqrt{n}$ and Tracy–Widom fluctuations $\simeq 0.90\, n^{1/6}$ [8], so $\delta\tau/\tau \simeq 0.45\, n^{-1/3}$. Our simulations reproduce both (`sims/causal_diamond.py`). Equivalently $\tau^2 = 2V$, where the volume V of the interval is its number of events divided by ρ. ours (interpretation) Geometry measured by counting is intrinsically fuzzy. The fuzziness is a relation between events, not a smearing in a background space.

3.2 Space: causal overlaps and a rectangle identity

literature For two events a, b with neither before the other, Boguñá and Krioukov [9] pick an event c in their common past and compare the intervals $I(c,a)$ and $I(c,b)$:

$$A = I(c,a) \setminus I(c,b), \quad B = I(c,b) \setminus I(c,a), \quad C = I(c,a) \cap I(c,b), \qquad O = \frac{V_C}{\min(V_A, V_B) + V_C}. \tag{4}$$

With c chosen at equal proper time τ_c from a and b, in $1+1$ dimensions $d = \tau_c(1-O)/\sqrt{O}$. We re-derived this.

ours In $1+1$ dimensions the choice of c does not matter. Step by step (Fig. 1):

N	$d < 2\ell$	2ℓ–10ℓ	$> 10\ell$
1000	23%	6.5%	5.2%
2000	15%	5.0%	3.4%
4000	19%	5.1%	2.5%

Table 1: Median relative error of Eq. (5) with the deepest c, flat $1+1$ sprinklings, 4 seeds (`sims/s4a_overlap_distance.py`). There is no significant bias above 2ℓ. The large-distance error falls roughly as $N^{-1/2}$. Below 2ℓ the error stays at $\sim 20\%$: this is the discreteness scale.

1. In light-cone coordinates $dt\,dx = du\,dv/2$, and for spacelike events $d^2 = \Delta x^2 - \Delta t^2 = -\Delta u\,\Delta v$.
2. With c at the origin, $I(c,a) = [0,u_a] \times [0,v_a]$ and $I(c,b) = [0,u_b] \times [0,v_b]$ are rectangles. Take $u_a < u_b$, $v_a > v_b$.
3. The rectangle from c to the join (u_b, v_a) splits into $C = [0,u_a] \times [0,v_b]$, $A = [0,u_a] \times [v_b,v_a]$, $B = [u_a,u_b] \times [0,v_b]$ and $D = [u_a,u_b] \times [v_b,v_a]$.
4. Opposite tiles have equal products: $C \cdot D = u_a v_b\,\Delta u\,\Delta v = A \cdot B$.
5. D has (u,v)-area $|\Delta u\,\Delta v| = d^2$, hence volume $d^2/2$.

Therefore, for *any* c in the common past,

$$d^2 = \frac{2\,V_A V_B}{V_C} \approx \frac{2\,N_A N_B}{\rho\,(N_C+1)}. \tag{5}$$

The strips A and B are long and thin, so they hold many events even when $d \sim \ell$; this is also why the Boguñá–Krioukov estimator works at short distances. Their equal-time choice is the special case $V_A = V_B$, and their $1+1$ formula is this identity in that case. Hu, Meyer and Meyer give a different exact $1+1$ formula, using a past and a future event at equal depth [45]. The identity is elementary and may exist in passing; we did not find the general-c product form among the papers citing [9]. The identity holds only in $1+1$ dimensions, because in more dimensions the overlap of two diamonds is not a diamond. The counts N_A, N_B, N_C come from disjoint regions and are therefore independent Poisson variables. $N_A N_B$ is then unbiased, but $1/N_C$ is not; $1/(N_C+1)$ has mean $(1-e^{-\lambda})/\lambda$ and removes the bias. Before this fix, a flat control read $+11\%$.

3.3 Curvature: the estimate drifts with depth

literature Curvature can be read from counts through the Benincasa–Dowker operator [10, 11]. In our runs (session 2) it gave the right sign and order of magnitude for de Sitter, but was noisy at 10^4 events. Related continuum results give the curvature corrections to the volume of a small causal diamond [12] and to chain counts inside it [13].

ours The rectangle identity provides a second route. Two-dimensional de Sitter space, $ds^2 = (H\eta)^{-2}(-d\eta^2 + dx^2)$, has the same light cones as flat space, so A, B, C, D are the same rectangles, but volumes are weighted by $dV = 2\,du\,dv/(H^2(u+v)^2)$. Integrating over a rectangle gives a logarithm of the conformal times at its corners,

$$V_\square = \frac{2}{H^2} \ln \frac{\eta_L\,\eta_R}{\eta_{\text{bottom}}\,\eta_{\text{top}}}, \tag{6}$$

and $C \cdot D = A \cdot B$ no longer holds exactly. For a, b on one slice and c a proper time τ_c below them, expanding for small d and then small τ_c gives

$$\frac{d^2_{\text{est}}}{d^2} = 1 - \frac{(H\tau_c)^2}{8} + \cdots = 1 - \frac{R\,\tau_c^2}{16} + \ldots, \qquad R = 2H^2. \tag{7}$$

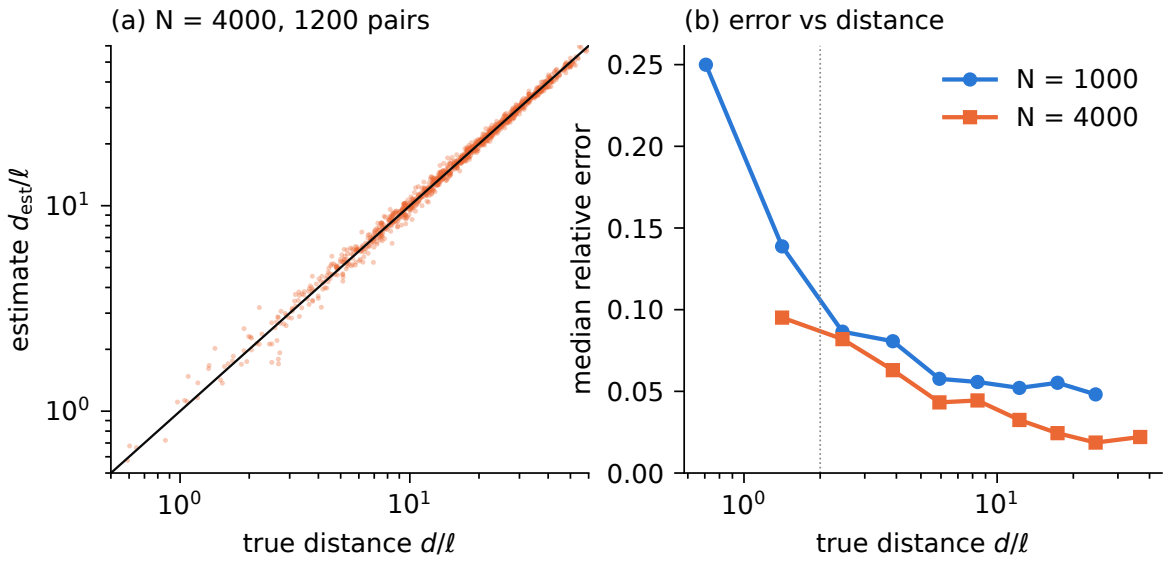


Figure 2: Eq. (5) with the deepest c on flat $1+1$ sprinklings, using only the causal order (4 seeds, 300 pairs each). (a) Estimate against true distance, $N = 4000$. (b) Median relative error per distance bin; the dotted line marks 2ℓ. Below 2ℓ the error is set by discreteness; above it, it falls with N (`sims/fig_part1.py`).

In flat space every c gives the same answer. In curved space the answer depends on how deep c lies, and the size of that drift is the curvature. We checked the coefficient $1/8$ numerically from the exact volumes (0.1252 at small τ). In sprinklings ($\rho = 10^4$, test events inserted by hand 14ℓ apart), the measured ratio agrees with the continuum in every depth bin for 200 fresh seeds, and a flat control shows no drift. Fitting $d^2_{\text{est}} = \alpha + \beta\tau^2$ with τ also from counts gives $R = -16\beta/\alpha$:

ρ	flat (true $R = 0$)	de Sitter (true $R = 2$; continuum fit in this window 2.13)
4000	-0.7 ± 0.6	1.3 ± 0.7
10^4	0.3 ± 0.4	1.6 ± 0.4

Figure 3 shows the measured drift against the continuum computed from the exact volumes for the same events c.

It is right on average, but a single pair is useless: the run-to-run spread is about 4, so about 100 pairs are needed for ± 0.4. A citation search (all papers citing [9]) found no curvature version of the overlap estimator; Boguñá and Krioukov only remark that their method extends locally to curved spacetimes. Other counting routes to curvature exist: chain abundances [13] and, recently, Ollivier–Ricci curvature [46].

3.4 Shape: an embedding that doubles as a manifold detector

literature Given squared intervals between M events, classical multidimensional scaling (MDS) with a Lorentzian Gram matrix reconstructs coordinates. Clough and Evans did this for causal sets, with longest chains for timelike pairs and the naive spatial distance for spacelike ones, and proposed the eigenvalue spectrum as a test of dimension and manifoldness [47]; see also [48, 14]. With $s_{ij} = (x_i - x_j)^T \eta (x_i - x_j)$, double centring $G = -\frac{1}{2} JSJ$, $J = I - \mathbf{1}\mathbf{1}^T/M$, gives $G = X\eta X^T$. A $1+1$ Lorentzian set shows one large negative eigenvalue (time) and one large positive one (space), with the rest near zero, and the eigenvectors give coordinates up to a Lorentz transformation.

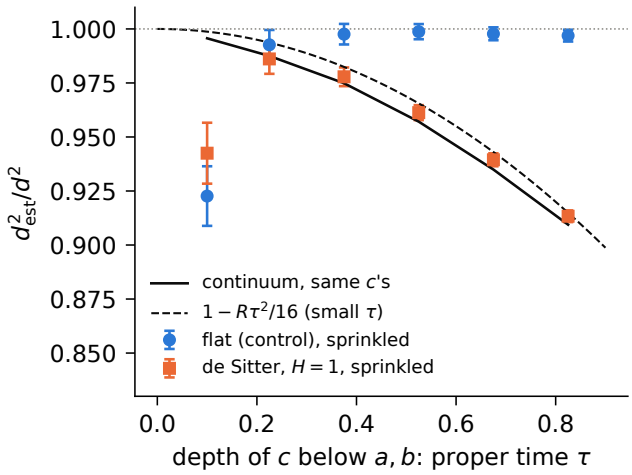


Figure 3: Depth drift of the rectangle estimate (ours). Two events a, b inserted by hand $\approx 14\ell$ apart; every sprinkled c in their common past gives one estimate; mean per depth bin over 400 sprinklings ($\rho = 10^4$), errors from the seed-to-seed spread. Solid line: continuum ratio from the exact de Sitter volumes, averaged over the same c's; dashed: the small-τ expansion, Eq. (7). The flat control stays at 1. The shallowest bin reads low in both geometries: there the tiles hold few events and the estimator is biased. Neighbouring bins share events and are ≈ 0.9 correlated, so the error bars are not independent.

ours With $s = -2N(I)/\rho$ for timelike pairs and Eq. (5) for spacelike pairs (all from counts), on $M = 150$ events of a flat sprinkling ($N = 2000$):

- positions are recovered to $0.43 \pm 0.07\,\ell$ across a region about 38ℓ wide, better than one event spacing;
- the map to the true coordinates is a Lorentz transformation to within 3–10%;
- the spectrum is $\lambda_- \approx -2.2$, $\lambda_+ \approx 2.2$, with all others below 0.1;
- a $2+1$ sprinkling fed through the same $1+1$ pipeline shows *two* large positive eigenvalues: the spectrum counts spatial dimensions even with the wrong distance formula.

Following [47], we turn this into a detector, and what we add is a calibration. It uses four intrinsic numbers: w_2, the share of the spectrum in one time and one space direction; $r_\pm$, the ratio of the second to the first eigenvalue of each sign; and the *stress*, the rms misfit of the best $1+1$ Minkowski embedding. Two calibrations matter. (i) Size: on flat universes, stress and r_+ are 0.22/0.42 at $N = 20$, 0.33/0.41 at 50, 0.20/0.16 at 100, 0.11/0.08 at 200 and 0.08/0.05 at 400. The detector is reliable only from $N \approx 100$–200; below 50 a flat universe looks like a crystal. (ii) Curvature: on de Sitter diamonds of fixed $N = 400$, the stress stays at the flat value up to $H\tau \approx 1$ and rises to 0.2–0.25 at $H\tau = 4$. So the embedding answers "is it a sheet?", while Eq. (7) measures curvature. (iii) Phases, and a blind spot we fixed. On the $N = 400$ Monte Carlo universes of Sec. 4 the detector as above passes every continuum-like state and fails the 4- and 7-layer crystals, but the $\sim$ 10-layer states are *not* reliably flagged: with $M = 40$ one of 7 sits in the flat band and two more just above it, and with $M = 80$ two others read stress 0.03, flatter than flat. Two things go wrong. The sample takes the events with the largest pasts, which in a layered order are the top layer or two. And in a layered order most sampled pairs have *zero* squared interval (linked pairs, or spacelike pairs with an empty strip), and MDS "fits" an all-zero matrix perfectly; a bilayer gives exactly that. The fix is a second number, the share of sampled pairs with zero length, taken over the top sample and over a sample spread evenly in height (`sims/r8_detector_spread.py`). It does not respond

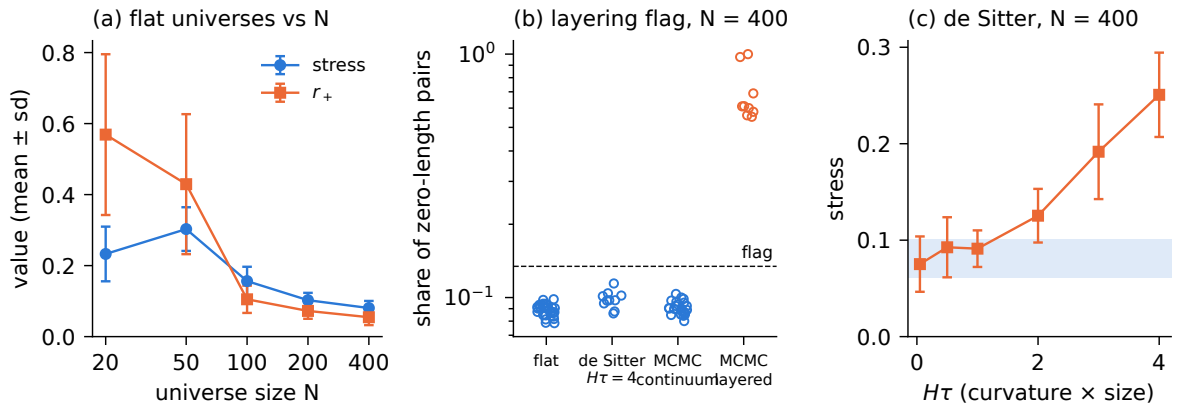


Figure 4: Calibrating the embedding detector ($M = 40$ in (a) and (c), $M = 80$ in (b)). (a) Flat universes (random 2D orders, 10 per size): the misfit (stress) and r_+ settle only for $N \gtrsim 100$. (b) The layering flag at $N = 400$: share of zero-length pairs for flat universes, de Sitter diamonds at $H\tau = 4$, and the final states of the Monte Carlo chains of Sec. 4 (continuum-like, 30–39 layers; layered, 4–11 layers). The dashed line, 1.5× the flat median, is the flag. The 13 states with 2–3 layers are not shown: the pipeline cannot run on them. (c) de Sitter diamonds of fixed $N = 400$ (8 per height): the stress leaves the flat band (shaded, ±1 sd) only for $H\tau \gtrsim 2$.

to curvature (de Sitter at $H\tau = 4$ reads like flat), and it separates layering by a wide margin (Fig. 4b): 0.075–0.103 on 100 flat universes and 0.080–0.103 on the 22 continuum-like states, against 0.55–1.0 on the layered ones, at $N = 400$. We flag a universe when the share exceeds 1.5× the flat median at the same N. We designed this after seeing the $N = 400$ states, so we tested it on fresh $N = 200$ chains: continuum-like states ($\beta \leq 0.3$) 0.12–0.13, all passing; layered ones ($\beta \geq 0.6$, 8–13 layers) 0.39–0.48, all flagged; flat 0.10–0.15 against a threshold of 0.18. Classically grown universes (0.17–0.19 at $N = 200$) sit near the threshold but fail on stress. literature The zero share is a cousin of the interval-abundance tests of manifoldness [51]; what we add is that the embedding detector needs it, and its calibration. The 2- and 3-layer crystals still break the pipeline (too few events with a common past), which we count as a fail. So the detector now asks two questions: *is it layered?* (zero share) and *is it a flat sheet?* (stress, which also responds to curvature).

4 Which universes are spacetime?

4.1 Classical growth fails

literature In sequential growth [15] events are born one at a time, and each newborn picks causes among existing events. ours We ran two rules: fixed link probability p (a steady state of fixed width) and $p_n = 3/n$ (an expanding universe from a single first event; layers of $1, 2, 3, 3, 5, 7, \ldots, \sim 280$ events). The second rule uses the birth order as a hidden clock, which C2 forbids; the covariant route is through cosmological renormalisation of p [16, 17]. Neither rule gives a $1+1$ spacetime. Their Myrheim–Meyer dimension drifts with scale, falling to ~ 1.2 for fixed p (a thin tube) and rising from 1.6 to 2.8 for $p = 3/n$. On the detector at $N = 100$ they fail every number: $w_2 \approx 0.33$ against 0.65 for the calibration, and stress ≈ 0.65 against 0.19.

4.2 Weights $e^{-\beta S}$: a continuum phase and a crystal

literature Weighting $1+1$ orders by $e^{-\beta S}$, with S the Benincasa–Dowker action, gives a phase transition between a continuum-like phase and a layered "crystal" [18, 19, 20]. ours We reproduce it ($\beta_c \approx 1.2, 0.7, 0.5$ for $N = 30, 50, 70$, at non-locality parameter $\epsilon = 0.21$). An $O(N^2)$ Metropolis update (a swap changes only two rows and columns of the causal matrix, so $K' = CC + C'E + EC$ with thin E) brings $N = 200$ to 0.03 s per sweep. Two clocks, height and past volume, give the same cosmic ordering in both phases, except for pairs within one or two layers (C5).

$N = 200$	ordering fraction	layers	w_2	r_+	stress
flat calibration	0.50	24	0.75	0.07	0.10
$\beta = 0.05$ (continuum)	0.51	24	0.74	0.10	0.11
$\beta = 0.6$ (crystal)	0.59	9	0.82	**0.39**	**0.28**

The continuum phase is indistinguishable from flat spacetime. The crystal asks for an extra spatial direction and fits badly.

$N = 400$: hot and cold starts never meet. ours At $N = 400$ we start each chain either hot (a random order) or cold (a two-layer order): 80 chains of 1.5×10^4 sweeps on the cluster, 4 independent seeds per (β, start), all complete for $\beta \geq 0.05$. Late-time layer counts (mean over seeds; seed spread ≤ 1):

β	0.05	0.08	0.12	0.2	0.3	0.45	0.7
hot start	35.4	34.9	34.3	33.3	31.9	10.4	10.0
cold start	35.3	5.3	3.4	2.1	2.0	2.0	2.0

Hot chains leave the continuum only at $\beta \geq 0.45$ and then stick in a ~ 10-layer state ($S/N \approx -12$); cold crystals melt only at $\beta \leq 0.05$ and otherwise thin to two layers ($S/N \approx -34$). *Caveat:* the cold chains accept $\leq 0.1\%$ of proposed moves, so "never meet" is partly a frozen sampler, not only metastability. The transition is therefore only bracketed, $0.05 < \beta_c(400) < 0.45$. literature This is consistent with the known first-order transition: the finite-size fit of Glaser, O'Connor and Surya at the same $\epsilon = 0.21$, $\beta_c \approx 1.66/(\epsilon^2 N)$ [20], gives 0.094 at $N = 400$, inside our bracket, and the symmetric bilayer order is known as the ground state [49]. The $N = 200$ "crystal" above is a metastable intermediate. Locating β_c needs parallel tempering.

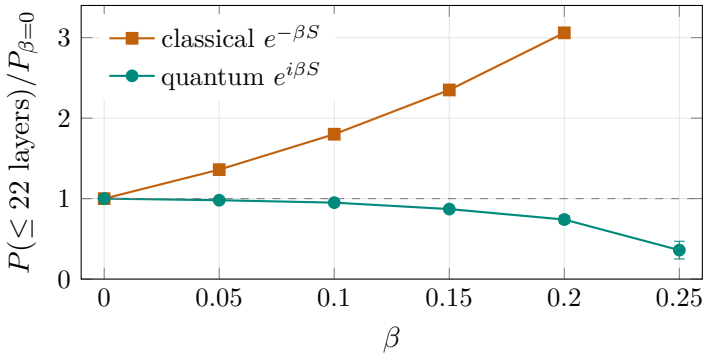
4.3 Quantum weights $e^{i\beta S}$

Records. ours (lesson from session 2) Phases only matter if different histories end in the same outcome. If every universe is its own outcome, $|e^{i\beta S}|^2 = 1$ and nothing interferes. We therefore choose a *record*, a relational property shared by many universes, and let universes with the same record interfere: $P_q(r) = |A_r|^2 / \sum_{r'} |A_{r'}|^2$, $A_r = \sum_{\text{record } r} e^{i\beta S}$.

A bound. ours (likely known) Since $|e^{i\beta S}| = 1$, $|A_r|$ is at most the number of universes with record r. In the quantum sum, rare universes such as crystals can only win if the common, manifold-like ones cancel each other out. With $e^{-\beta S}$, rare universes with low action win by sheer weight.

Exact sums, $N \leq 9$. ours Summing over all $N!$ orders with record = number of layers, $e^{-\beta S}$ collapses onto flat layered universes ($N = 9$, $\beta = 8$: 2 layers with 76%), while $e^{i\beta S}$ suppresses them by destructive interference (2 layers: 0%; 5 layers: 62%). Such universes are too small to say whether they are spacetime.

Sampling, $N = 200$. ours The amplitude of a record is proportional to the average of $e^{i\beta S}\mathbf{1}_r$ over uniformly random orders, which can be sampled. This works while the average phase, $\approx e^{-\beta^2\sigma_S^2/2}$, is not tiny. At $N = 200$, $\sigma_S = 12.0$, so the average phase is 0.49 at $\beta = 0.1$ and 0.06 at $\beta = 0.2$. We drew 10^6 orders on the cluster (50 independent seeds; errors are jackknife over seeds). The record is the layer count (the height clock of C5), grouped into bins. The probability of the few-layer bin (≤ 22 layers; the mean is 24.6), relative to its value at $\beta = 0$:



Classically, raising β drives the universe toward few layers, the approach to the crystal. Quantum weights do the opposite and suppress few-layer universes. This agrees with the exact sums and with Loomis and Carlip [24], who show analytically that e^{iS} suppresses two-layer orders. (The classical reweighting is reliable up to $\beta = 0.2$, with an effective sample size of 1,022.)

But the mechanism is simple. Within each record the action is close to Gaussian, with spread σ_r. Then $|A_r| \approx n_r e^{-\beta^2\sigma_r^2/2}$, so

$$P_q(r) \propto n_r^2\, e^{-\beta^2\sigma_r^2}, \tag{8}$$

and this reproduces the sampled numbers to 0.001 for $\beta \leq 0.15$. Few-layer universes have a slightly larger spread ($\sigma = 11.73$ against 11.48), so their phases cancel more. In the range that sampling can reach, "quantum suppression" is a variance effect, not a structured interference pattern. literature This is the small-phase form of the mechanism in the analytic treatments [24, 50], where steepest descent on binomial sums over link numbers gives the suppression; what the sampling adds is that, at $N = 200$ and reachable β, nothing beyond it is visible. With record = stress quartile, quantum and classical weights drift by the same small amount (Q1: 0.261 ± 0.003 against 0.258 at $\beta = 0.15$). A 1,600-sample pilot that suggested a large shift toward manifold-like universes was noise. Uniform sampling is exhausted at $\beta \approx 0.2$ for $N = 200$; real interference structure needs exact or layer-by-layer methods.

5 A first actor: a particle as a sum over chains

An event of the model is an atom of spacetime volume, about a Planck 4-volume, not a laboratory event: a femtometre-scale collision spans $\sim 10^{79}$ of them. Particles have to live *on* the events. literature A classical free particle maximises proper time, so it is the longest chain [7]. Johnston's quantum particle [27] sums over all chains, with amplitude a per hop and b per stop: with $\Phi = aC$ (C the causal matrix),

$$K = \Phi + b\Phi^2 + b^2\Phi^3 + \cdots = \Phi(I - b\Phi)^{-1}, \qquad a = \tfrac{1}{2},\ b = -m^2/\rho \ \ (1+1). \tag{9}$$

Step by step: in $1+1$ the expected number of chains $x \prec z_1 \prec \cdots \prec z_k \prec y$ in an interval of area $\tau^2/2$ is $(\rho\tau^2/2)^k/(k!)^2$, so $\langle K\rangle = \frac{1}{2}\sum_k(-(m\tau/2)^2)^k/(k!)^2 = \frac{1}{2}J_0(m\tau)$, the retarded Klein–Gordon propagator. We reproduce it on sprinklings.

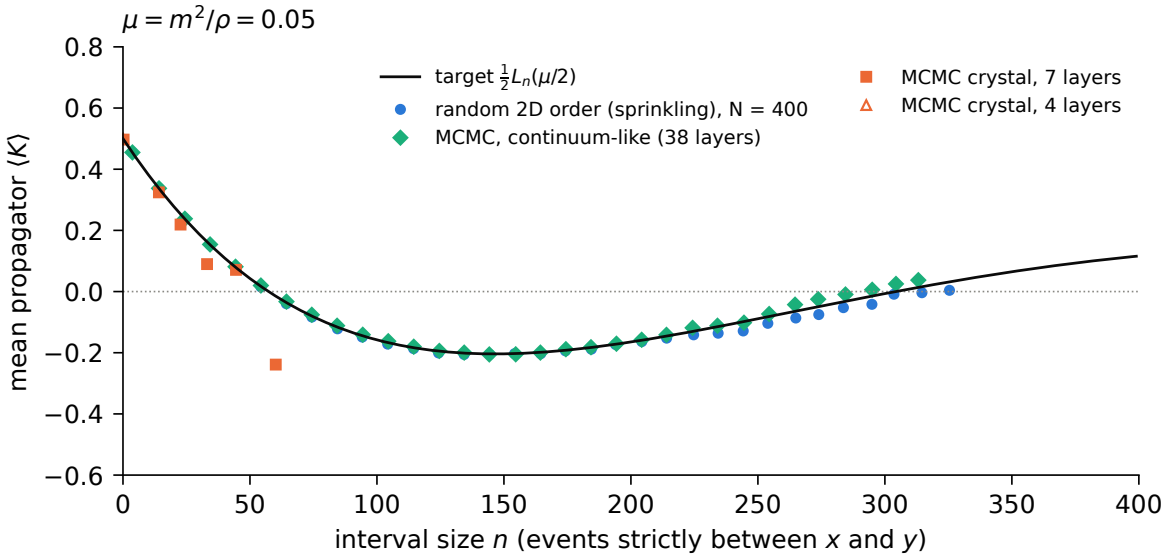


Figure 5: Johnston's propagator averaged over related pairs with the same interval size n, against the density-free target $\frac{1}{2}L_n(\mu/2)$, at $\mu = 0.05$ and $N = 400$. A random 2D order (a flat sprinkling) and a continuum-like Monte Carlo universe follow the target; at large n few pairs remain and the bins get noisy. A 7-layer crystal falls off it; in a 4-layer crystal every interval has fewer than 10 events, so it gives a single point near $n = 0$ and the mass never shows (`sims/fig_part1.py`).

An exact target from the order alone. ours K depends only on $\mu = m^2/\rho$: $K(x, y) = \frac{1}{2}\sum_k c_k(x, y)(-\mu/2)^k$, with c_k the number of k-element chains inside the interval. In a uniformly random 2D order, the n events of an interval are again a random 2D order, and k of them form a chain with probability $1/k!$, so $E[c_k \,|\, n] = \binom{n}{k}/k!$ [28] and

$$E[K \,|\, n] = \tfrac{1}{2}\sum_k \binom{n}{k}\frac{(-\mu/2)^k}{k!} = \tfrac{1}{2}L_n(\mu/2) \;\approx\; \tfrac{1}{2}e^{\mu/4}J_0\big(\sqrt{2\mu(n+\tfrac{1}{2})}\big), \tag{10}$$

a Laguerre polynomial in the interval size. Averaging over Poisson n with mean $\rho\tau^2/2$ gives exactly $\frac{1}{2}J_0(m\tau)$, so Johnston's result is the Poisson average of this one, which needs no density at all. literature The identity itself is combinatorics: summed over all permutations at $x = 1$ it is OEIS A002720, $n!\,L_n(-1)$, the total number of increasing subsequences [55]. ours Its application to the propagator was not found in any of the 99 papers on arXiv that cite Johnston's propagator papers [27, 56], searched in full text; recent path-sum work averages over Poisson sprinklings throughout [57, 58]. Expanding in μ, the first order that can fail is the ordering fraction, i.e. the Myrheim–Meyer dimension: the propagator is a resummed tower of dimension tests.

Tests. ours Continuum-like $N = 400$ universes match the target as well as random controls do (rms 0.013–0.020 at $\mu = 0.05$; Fig. 5). Crystals fail: a 4-layer one has no interval larger than 9 events, so K never turns negative and "mass" means nothing; a 7-layer one falls far below the target (too many chains). All classically grown universes ($N = 1500$) fail systematically, and one ($p = 0.02$) runs away to $|K| \approx 10$, twenty times the 2D bound $\frac{1}{2}$. In curved 2D a term $(R\tau^2/48)J_2(m\tau)$ appears [29], derived from chain abundances. *At fixed n* curvature shows up too: in a 2D de Sitter diamond conditioned on n events, the order depends only on $(n, H\tau)$, and the share of related pairs rises by $\approx 0.027\,(H\tau)^2$ for every n we tried (10–60). But the propagator is no more sensitive than that share alone: a 3σ detection needs the same

number of intervals (e.g. 175 vs 175 at $n = 30$, $H\tau = 1$). So the fixed-n propagator measures curvature through the known chain-abundance route [13], not a new one. Limits: one scalar, retarded propagator only, no backreaction, and an exact target only in 2D. Interactions are open: Johnston considered summing vertices over events and set it aside as ambiguous.

Part II: what must a growing universe remember?

6 Does quantum growth have a stable past?

One amplitude per birth. literature In sequential growth every causal set can be grown by adding events one at a time, each newborn taking any down-set of the existing events as its past [15]. ours A birth f changes the Benincasa–Dowker action by

$$\Delta S_f = 4\epsilon - 8\epsilon^2 \sum_{x<f} f(n_{xf}, \epsilon), \tag{11}$$

with n_{xf} the number of events between x and f (summing over births reproduces S to 10^{-10}). This is just the newborn's own term in the retarded operator that defines the action [10], and the number of small intervals per element grows like $\ln N$ [51]. Give each birth the amplitude $e^{i\beta\Delta S_f}$. The amplitude of a causal set C is then (number of natural labellings of C) $\times e^{i\beta S(C)}$: covariant, with the counting measure supplied by the growth itself. literature That a sum over labelled causal sets weights each one by its number of labellings is noted in [3, §6.4]. Summed exactly to $N = 8$ (2.8×10^6 growth paths): at $\beta = 0$ a three-layer, Kleitman–Rothschild-like sector carries 80% of the weight; a window $\beta\epsilon^2 \approx 0.3$ suppresses it by interference, as known [24], but what is left is not yet spacetime-like.

No normalization; does the past stabilise? ours Classical growth normalizes the transition weights at every step. Doing the same with amplitudes breaks covariance, and the violation grows with N (a factor of ~ 10 at $N = 22$). We drop it and instead ask that the past stabilise as the universe grows. For a grown stem X define the *future factor*

$$F_N(X) = \sum_{\text{continuations } X \to N} e^{i\beta(S_N - S_X)}. \tag{12}$$

The amplitude of a question about the past is $\sum_X e^{i\beta S_X} F_N(X)$ over the stems X that answer it, so its probability settles as $N \to \infty$ only if the ratios $F_N(X)/F_N(Y)$ converge: the future must factorize. Stems are the right observables, since stem sets generate the covariant questions classically [31].

literature Every quantum growth model we found imposes a sum rule on the transition amplitudes [52, 30, 53], which makes $F_N = 1$ by construction and turns the problem into extending the quantum measure; Sorkin proposed defining measures as limits along sequences of events [54]. The closest open problem stated in the literature is that the space of quantum growth models and the truncated decoherence functional are "largely unexplored" [3, §6.4]. A citation search found no computation of F_N.

Exact future factors to $N = 11$. ours Memoizing unlabelled posets (2.77 million at $N = 11$, OEIS A000112) gives F_N exactly (47 minutes on one cluster core). At $\beta = 0$ the ratio of the 2-chain stem to the 2-antichain stem is exactly $3/(N+1) \to 0$: the layered sector wins, and "the first two events are unrelated" becomes certain. To see what the phases do beyond counting, we divide by this: $q_N = r_N(\beta)/r_N(0)$, with $r_N = F_N(X)/F_N(\text{antichain})$.

q_N, $\epsilon = 0.21$	$N = 8$	9	10	11
2-chain stem, $\beta = 7$ (window)	$2.25\angle{-0.99\pi}$	$1.91\angle 0.98\pi$	$1.89\angle 0.99\pi$	$1.86\angle{-0.98\pi}$
3-chain stem, $\beta = 7$	$28.5\angle 0.29\pi$	$9.8\angle 0.58\pi$	$7.3\angle 0.55\pi$	$7.2\angle 0.68\pi$
2-chain stem, $\beta = 2$	$1.14\angle 0.17\pi$	$1.15\angle 0.19\pi$	$1.14\angle 0.20\pi$	$1.14\angle 0.22\pi$

Three behaviours. (i) Inside the suppression window the smallest stem converges: $|q_N| \approx 1.87$ with the phase locked at π, changing by 1–2% per added event. (ii) Larger stems still move: 3- and 4-element stems change by 20–40% per event in the window, and above it ($\beta = 12$) everything is erratic. (iii) At weak coupling the magnitudes have converged, but every stem's phase drifts *linearly* in N, by $\approx 0.008\pi\,\beta$ per relation in the stem per added event, for all 20 stems computed (slowly decreasing with N). Each relation in the past seems to shift the action of every later birth by a roughly fixed amount. Stems that answer the same question with different numbers of relations then keep a relative phase that rotates, and the probability of the question oscillates instead of settling. Whether the drift dies out fast enough matters: a drift per event falling like $1/N$ still adds up to a phase growing like $\ln N$.

Where the drift comes from. ours To first order in β, $F_N(X) \approx \mathrm{Count}_N(X)\, e^{i\beta m_N(X)}$, where $m_N(X) = \langle S_N - S_X \rangle$ is the mean action of the future, averaged over all growth continuations of X with equal weight. The drift per event is then βd_N, with d_N the change of $m_N(X) - m_N(\text{antichain})$ when one event is added. We compute m_N exactly (as the phase of F_N at $\beta = 10^{-4}$, divided by β). For the 2-chain at $\epsilon = 0.21$, $N d_N \approx 0.21$ at $N = 6$–10; the exact $\beta = 2$ data give 0.24, the 10% difference being the next cumulant, so this mechanism accounts for the drift. How fast d_N falls depends on the non-locality scale. Writing $d_N \propto N^{-a}$, from $N = 6$–10:

ϵ	0.21	0.35	0.5	0.7
effective exponent a	≈ 0.95	$1.5 \to 1.8$	$2.2 \to 2.6$	≈ 3.3

The weights $f(n, \epsilon) \sim (1-\epsilon)^n$ die as the number of events between a past relation and a newborn grows, so once N exceeds the non-locality scale the exponent passes 1 and the accumulated phase converges. At $\epsilon = 0.21$ that scale is $\sim 1/\epsilon^2 \approx 23$ events, beyond exact enumeration, and the exact $\beta = 2$ data turn over only at the last step ($N d_N = 0.246 \to 0.239$ at $N = 10 \to 11$). *Prediction for $N = 12$:* $N d_N \approx 0.233$ for the 2-chain at $\epsilon = 0.21$, $\beta = 2$; a flat or rising value would mean a $\ln N$ phase. So at weak coupling the drift is probably not fatal. Inside the window the first-order picture does not apply, and there the larger stems are still moving. Verdict: open, with a falsifiable next step.

7 How much past does the present need?

A cosmological clock (C5) suggests rewriting the sum over histories as a film: a state for "now" and an operator taking it to the next moment (C7). That needs the action to split into steps that each see only a small piece of the past. We measured how far that goes.

Thick height layers fail. ours Truncate S to pairs at most m height layers apart. Links skip layers: in continuum $1 + 1$ orders, 61% of links skip at least one layer at $N = 200$, and the longest spans 23 of about 24 layers. To keep the error below $0.25\,\sigma_S$ the memory must be $m \approx 9, 10, \gtrsim 14$ layers at $N = 100, 200, 400$, about 40% of the age. At $N \leq 9$ the truncated quantum sums deviate from the exact ones by up to 0.47 (total variation) when only 4.5% of the pairs are dropped. A volume clock gives different answers from the height clock, so a height truncation also breaks C5. (The crystal needs only $m \approx 4$.)

Growth and causal closeness. literature We let events be born one at a time, each with any set of existing events as its past but never in the past of an existing event: Rideout–Sorkin's internal temporality [15]. Every causal set can be grown this way, the dates (heights) of events never change, and $e^{i\beta S}$ does not depend on the birth order. ours In the birth step (11), keeping only x with $n_{xf} \leq 10$ leaves an error of 0.06–0.07 σ_S at $N = 100, 200, 400$, flat in N, and 0.07 in the crystal. The cut is a count, so it uses no clock. About 16, 22, 29 past events are kept per birth, growing like $\ln N$, but they span the whole age: the influence of the past is sparse and long-range in time, and local in "events in between".

The thick present. ours If f needs x, it also needs everything between them, so a state holding all events that some later birth still needs is self-contained. We measured its size in hindsight, which gives a lower bound:

universe ($1+1$, $n_{\text{cut}} = 10$)	state
diamond (open space)	0.6–0.8 N
static circle of length W	$\approx 0.5\,W^2$, independent of the age
decelerating closed FRW ($a \propto t^{1/2}$)	oldest needed event $\approx 0.5\,W$ of conformal time back
accelerating closed FRW (e^{Ht}, t^2)	0.9–1.0 of the past, never retired

An event stays needed until its light cone wraps around space, about $W/2$ of conformal time. The film therefore has a finite memory if and only if the universe has no event horizon. Our universe accelerates, so by this measure its present needs the whole past. The per-birth step is cheap; the storage is not.

Per observer: finite, but a volume law. ours The "whole past" of an accelerating universe is a stack of de Sitter static patches. Per patch, the events still needed are finite and stationary, so the horizon does bound the memory. But they scale with the spacetime volume: about 3ρ, 6.3ρ, 9.6ρ events in $d = 2, 3, 4$, with $\rho = 1/(\ell H)^d$ from 25 to 400 (exponent ≈ 0.97; an area law would give 0, $\frac{1}{3}$, $\frac{1}{2}$). The reason is Little's law from queueing: memory = birth rate $\times$ how long each event stays needed. The second factor, 1–3 Hubble times, does not depend on ρ, because under C4 an event cannot be forgotten until its light cone leaves the patch. In $3+1$ with $\ell \approx \ell_{\text{P}}$ this is $\sim (L_H/\ell_{\text{P}})^4 \approx 10^{244}$ events per patch, against a de Sitter entropy of $\sim 10^{122}$ (a comparison of scales, not a match). literature The numbers are Sorkin's: $N \sim 10^{240}$ elements to our past and $\Lambda \sim N^{-1/2}$, so a horizon entropy $\sim 1/\Lambda \sim \sqrt{N}$ [59]. What we add is the reading of N as *memory* the growth must hold, the mechanism (Little's law, residence time set by C4), the per-patch accounting, and the criterion "finite iff no event horizon". Bousso's N-bound reaches finiteness from the same causal-diamond logic, for entropy rather than growth [60]. Either the count overstates what the universe must hold (a clumsy description of a simpler state), or the dynamics is missing something.

A ladder. ours Three levels of description, each needing less memory:

level	the future needs	memory scales as	status
events	the past events newborns can link to	spacetime volume	measured (above)
geometry and fields	data on one slice (an initial-value problem)	spatial volume	established for GR; open for us
area	slice data minus what gravity forbids	horizon area	needs gravity

Coarse-graining into cells of a fixed size ξ keeps a volume law with a smaller coefficient. An area law in $3+1$ would need $(L/\xi)^4 = (L/\ell)^2$, i.e. $\xi = \sqrt{L\ell}$, a cell growing with the universe. This is the UV/IR scale of Cohen, Kaplan and Nelson [33], near the dark-energy length; we note it and do not lean on it (any combination of L and ℓ_{P} is a power of 10^{61}).

The second rung fails for the causal-set operator. ours Put a test event at the top of a flat $1+1$ diamond and compute what it feels: Sorkin's operator on $\phi = 1$, raw and smeared ($\epsilon = 0.1,\ 0.03$) [5]. Then shuffle the events inside each cell of a light-cone grid, keeping the count per cell, and recompute. The share of the variance created by shuffling, $f_{\text{micro}} = E[\mathrm{Var}(b\,|\,\text{cells})]/\mathrm{Var}(b)$, is 1 if the cell counts explain nothing and 0 if they suffice (means ≈ 0 in all runs; 16–24 independent bases):

events per cell	raw ($N = 500/2000/8000$)	smeared $\epsilon = 0.03$ ($N = 500/2000/8000$)
≥ 31	1.0/1.0/1.0	1.0/0.90/0.99
2	0.99/1.07/0.95	0.20/0.30/0.66
0.5	0.92/0.94/–	0.07/0.12/–

The raw operator has no macrostate at all: moving events within cells holding half an event already scrambles it. The smeared one admits one only at $\lesssim 2$ events per cell, and that gets worse as the universe grows (cells tilted by a boost are worse still). What a newborn feels comes from a thin sliver along its light cone that stretches over the whole past, which no fixed cell resolves. Classical coarse-graining compresses by at most a small factor. literature In substance this is Sorkin's observation that the fluctuations of the discrete $\Box$ grow with N and sit along the light cone [5]; the shuffling test makes it quantitative for memory. Caveats: one newborn, flat space, cells tied to one frame.

Quantum memory as a matrix. ours Let p be the universe after k births (unlabelled) and F the final universe, and let B_{pF} be the sum of $e^{i\beta\Delta S}$ over all growths from p to F; its row sums are the future factors F_N. Pasts with equal rows can share a classical memory slot, so the classical memory is $\log_2$ of the number of distinct rows; linearly dependent rows can share quantum memory, so the quantum memory is $\log_2 \operatorname{rank} B$ (the bond dimension of a sequentially generated state [34]; quantum models can need less memory than classical ones [35], and the optimal ones are matrix product states whose memory is the past/future entanglement [61]). These tools have not, as far as we found, been applied to causal-set growth. A structural fact follows from Eq. (11): ΔS depends only on the endpoints, so $B = \mathrm{diag}(e^{-i\beta S_p})\,\mathrm{Count}\,\mathrm{diag}(e^{i\beta S_F})$, and *the phases cannot change any memory measure* when the outcome is the whole final universe. They can only matter for coarser outcomes. With a coarse record of F (height, minimal and maximal events, links; $N = 7$, $k = 5$, 63 pasts) the effective memory (entropy of normalized singular values) is 3.78 bits at $\beta = 0$, 4.05 at $\beta = 7$, 4.45 at $\beta = 12$ and 5.34 for random phases: interference adds memory.

Records. ours One motivation (Roberto's) is that links might stay undetermined precisely because nothing stores them, with quantum mechanics as the universe's way of saving memory. Ideas deriving quantum theory from limited information exist [36, 37], and finite information density has been argued to imply indeterminism, even classically [62]; a deeper search found none deriving interference from a storage limit of the universe. Two corrections sharpen it: storing a probability distribution or a wavefunction costs more than storing one state, so any saving must come from *summing* over what is never determined; and the saving needs one real history, since a universe holding all branches stores more, not less. We adopt Sorkin's quantum measure theory [38]: unchanged dynamics, one real history (made precise by co-events [63]), and probabilities only for alternatives that no longer interfere. A record is then an alternative that can never interfere again, which needs no threshold, unlike physical collapse models, which need a rate: relativistic GRW [39], or gravity-related Diósi–Penrose collapse, whose parameter-free form is excluded [40]. Two findings. (i) If every event saw its whole past, everything inside any future event's past would be a record and the memory a volume law again; any saving needs events that see their past only partially. (ii) Taking as the screen what the last-born

event sees (its full past, its size and number of links, or only the number of links), with the universe after k births as the alternatives, the mean coherence between alternatives at $N = 8$, $k = 5$ is:

screen	$\beta = 0$	$\beta = 8$	random phases
full past	0.55	0.21	0.15
size, links	0.75	0.38	0.22
links	0.97	0.72	0.49

No pair is ever exactly decohered. The coarser the screen, the more stays undetermined; that saving is classical ($\beta = 0$ has no interference). The phases make alternatives *more* distinguishable: here quantum mechanics makes records rather than saving memory. Caveats: $N \leq 8$, alternatives labelled by birth order rather than covariant stems, one observer. literature Both findings have relatives. That records cost storage matches Halliwell's conjecture that the information needed to describe decoherent histories is about what is lost to the environment [64]. That no pair ever decoheres exactly matches, in spirit, Bousso and Susskind's point that a finite causal diamond has a finite Hilbert space, so decoherence is never exact [65]. In holographic space-time each causal diamond carries $\log \dim \mathcal{H} = A/4$ [66], the closest existing framework to an area-law memory per patch. Collapse events on a causal structure have been modelled on a regular $1 + 1$ null lattice [67, 68]; nobody seems to have run them on random or grown causal sets, or counted records per region.

8 Discussion

Where this sits. The kinematics is causal set theory: links, infinite valency, Number = Volume, the longest chain as proper time, growth, the Benincasa–Dowker action, the phase transition, overlap distances and Lorentzian embeddings are theirs. For each item we once called ours, Table 2 gives its status after a citation crawl (all papers citing [9, 27, 56, 29, 42], full texts where available, plus targeted searches). "Not found" means not found in that crawl, not "new". On the dynamics side, a cosmological clock (C5) would license writing the sum over histories as a transfer matrix between cosmic-time layers, with operators that create events (C7). That puts us next to causal dynamical triangulations and Hořava–Lifshitz gravity, whose preferred foliation is fundamental rather than emergent; we must show that the difference is real.

Limitations.

1. *Everything is* $1 + 1$. The rectangle identity is $1 + 1$ only, and $1 + 1$ GR has no Newtonian limit, so C6 needs $3 + 1$.
2. $3 + 1$ *is not tamed*: 10^{124} neighbours per event.
3. *Matter is a probe, not a source.* One free scalar on a fixed order; no interactions, no backreaction, nothing gravitates.
4. *Small universes.* The detector needs $N \gtrsim 100$–200. Exact quantum sums stop at $N \approx 11$, and sampled ones are limited by the average phase.
5. *The quantum result is a variance effect.* Sampling reaches only $\beta \lesssim 0.2$ at $N = 200$. The record and ϵ are chosen, not derived.
6. *The* $N = 400$ *sampler freezes* in the layered phase; β_c is bracketed, not measured.
7. *No prediction.* In $1 + 1$ the cutoff is invisible to wave propagation, not merely far beyond tests.

8. *Memory.* Per patch it is a volume law. Neither cell coarse-graining nor quantum phases reduce it in what we computed. The quantum tests use $N \leq 8$ and birth-order alternatives, not covariant stems.

Waiting and next computations.

- F_N at $N = 12$ (about 5×10^7 unlabelled posets up to 11 elements, ∼110–125 GB): does the 2-stem stay converged, do 3-stems settle in the window, and does Nd_N fall to ≈ 0.233 as predicted?
- Parallel tempering at $N = 400$, to test the literature value $\beta_c \approx 0.094$.
- Records with covariant alternatives (stems) and events that see a fixed budget of their past; records per patch against births.
- The cutoff check, the overlap estimator and the detector in $2 + 1$ and $3 + 1$.

Unexplored. By us: matter as a source, interactions, the test that the matter frame is the cosmic frame (C5), $3 + 1$. By the literature, as far as we found: unnormalized quantum growth and the truncated decoherence functional ("largely unexplored" [3, §6.4]); collapse or co-event dynamics on random causal sets with records counted per region; the memory of covariant growth against the N-bound; computational-mechanics measures of memory [35, 61] for classical sequential growth; boundary-free (compact) regulators for link observables [43].

Two notes. Part I could become a short methods note (rectangle identity, depth-drift curvature, calibrated embedding statistics, Laguerre target), whose claims are checkable and modest. Part II is a research programme whose central result, the future factor, is still open; it should wait for $N = 12$ and an analytic handle on the drift.

A Paths explored and dropped

The full record, with dates and the test that killed each path, is kept in the notebook, in the file `GRAVEYARD.md`.

- **A regular crystal of events.** Polygonal light cones and a lattice rest frame [21, 22].
- **Quantum blur of positions.** Blur in *what*? It presupposes a background, and an equal superposition over rapidity cannot be normalised.
- **A quantum chain as the source of blur.** Background-dependent. The atomic-linewidth analogy was overstated: coupling *reduces* the blur.
- **Expansion as a source and gravity as a sink of points** (river model [41]). It needs an aether.
- **A link to Bell non-locality.** Links are timelike; Bell correlations are spacelike.
- $(L/\ell_{\mathrm{P}})^2 \sim 10^{122}$ **and the cosmological constant.** Numerology: every combination of L and ℓ_{P} is a power of 10^{61}.
- **The manifold detector on exact quantum sums.** At $N \leq 12$ the detector cannot tell a flat universe from a crystal; replaced by sampling.
- $\eta_{\max}$ **as a dispersion correction.** Killed in $1+1$ (Sec. 2). An intermediate version claimed a jump to $-1/\ell^2$ once a wavelength exceeds L; that was an artefact of a rapidity-wedge regulator.

- $\eta_{\max}$ **as the long-link suppression of Caravelli and Markopoulou [25].** A different object: a spatial lattice with external time, giving an energy-dependent mass.
- **A transfer matrix on thick height layers.** The memory must grow to $\sim 40\%$ of the age, and the truncation breaks clock independence (Sec. 7).
- **A finite-state film in an accelerating universe.** The thick present never retires its events (Sec. 7).
- **Quantum weights at $N = 100$ on stress quartiles.** Superseded by $N = 200$; the small-sample hint was noise.
- **Normalizing quantum growth at every step.** Breaks covariance, increasingly with N.
- **"Memory shared with the outside" as the area-law part of the memory.** It is all of it: outside events need old events forever.
- **A cell macrostate for what a newborn feels** (the second rung of the ladder). Shuffling within cells scrambles it, worse as N grows.
- **"Probabilities are cheaper to store than states."** The opposite; only summing over undetermined alternatives can save memory.
- **Diósi–Penrose collapse with the discreteness length as its size cut-off.** A smaller size means faster collapse; excluded by the underground bound.
- **Exactly orthogonal futures of birth-order pasts as records.** With an open future no record ever forms: any two small pasts fit in a common future.
- **Claims given back (session 10).** The MDS manifold test [47], the wedge -2 [42], the age of the universe as the IR regulator [5], 10^{244} vs 10^{122} [59] and the bilayer ground state [49] were in the literature before us.

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claim	status	closest prior work
Age of the universe as the IR regulator; cosmic frame	known	Sorkin [5]; frame from a cutoff [43]. Ours: $\eta_{\max} \approx 142$
Wedge gives -2; limits do not commute	known	[42]
Finite past diamond: $\Box$ exact up to e^{-V}	known (constant field)	per-event form of [44]; long waves agree with [23]
Rectangle identity, any c	not found	special case $V_A = V_B$ in [9]; other exact formula [45]
Curvature from depth drift	not found	chain abundances [13]; Ollivier–Ricci [46]
Lorentzian MDS spectrum as manifold test	known	[47]. Ours: count-based inputs, calibration vs N; the zero-length share it needs to see layering (a cousin of [51])
$N = 400$ hot/cold never meet	new data, known physics	first-order transition [20]; bilayer ground state [49]
$e^{i\beta S}$ suppression = action variance at reachable β	mechanism known	[24, 50]. Ours: the Monte Carlo check
Laguerre target $\frac{1}{2}L_n(\mu/2)$	identity known; application not found	[55, 28]; 99 citers of [27, 56]
Labelling weight of grown causal sets	known	[3, §6.4]
Future-factor criterion; exact F_N to $N = 11$	not found	sum-rule models [52, 30]; [54]
Weak-β drift = mean future action; decay faster than $1/N$ beyond the non-locality scale	not found	—
Propagator at fixed n as curvature probe	reduces to known	chain abundances [13]
Birth step cut $n_{xf} \leq 10$, error flat in N	not found	$\ln N$ small intervals [51]
10^{244} vs 10^{122}	known	[59]. Ours: memory reading, Little's law, "finite iff no horizon"
Cell coarse-graining fails	known in substance	[5]. Ours: the shuffling test
Endpoint phases are a gauge of B	elementary	not stated for causal sets
Records via screens; phases make records	not found	[64, 65]

Table 2: What is ours after the session-10 literature crawl.